

TCAN1051HVD



DevLab: TCAN1051HVD CAN Transceiver Module

Professional electronic component

v1.0
2025-10-02
Rev. A

PRODUCT OVERVIEW

The TCAN1051HVD is a high-speed CAN transceiver designed for automotive and industrial applications. It provides a robust interface between a CAN controller and the physical CAN bus, ensuring reliable data transmission in harsh environments. This module is ideal for use in electric vehicles, industrial automation, and other applications requiring reliable CAN communication. ### Quick Setup

PRODUCT WIKI

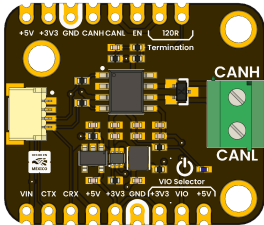
DATASHEET

BUY NOW

GETTING STARTED | Feature / Component | Details | | Communication | High-speed CAN interface for reliable data transfer. | | Operating Voltage | Supports 3.3V and 5V logic levels. | | Input Port | JST 1.0 mm 4-pin connector for power and CAN signals. | | IC1 – TCAN1051HVD | Automotive-grade CAN transceiver IC. | | IC2 – AP2112K | 3.3V LDO regulator for stable voltage. | | IC3 – TPS61023 | 5V boost converter for efficient power delivery. | | L1 – Power LED | On-board LED for power status indication. | | JP1 / JP2 | 2.54 mm castellated holes for flexible mounting or soldering. | | J1 – CAN Data Port | JST 1.0 mm connector for CAN bus data lines. | | J2 – Screw Terminal | Terminal block for secure CAN bus wiring. | | SB1 – Logic Voltage Selector | Solder bridge to select logic voltage (VIO). |

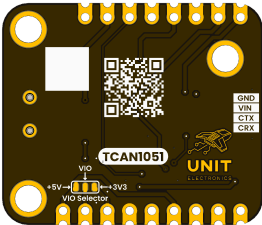
PRODUCT VIEWS

TOP VIEW



Component placement and connectors

BOTTOM VIEW



Underside components and connections

KEY TECHNICAL SPECIFICATIONS



CONNECTIVITY

Primary Interface:GPIO (Interrupt)

Connector Type:JST 4-pin 1.0mm

Logic Levels:VCC-referenced (2V – 5.5V tolerant)

PIN CONFIGURATION

FUNCTION		NOTES
Power	Supply	3.3V or 5V

COMPONENT REFERENCE

REF.	DESCRIPTION
IC1	CAN Transceiver
IC2	AP2112K 3.3V Regulator
IC3	TPS61023 5V Regulator
L1	Power On LED
JP1	2.54 mm Castellated Holes
JP2	2.54 mm Castellated Holes
J1	JST 1 mm pitch for CAN Data
J2	Terminal Screw Block for CAN Bus

KEY FEATURES

Touch-only sensing

No physical press required – reacts to proximity of a finger.

Auto-calibration

Compensates for environmental changes and drift.

On-board pull-up/down

Ensures clean digital output.

JST PH-2.0 connector

Quick-disconnect cable interface.

Fast response

< 80 ms touch detection time.

Selectable modes

Momentary or toggle output (via solder-jumper on the board).

Mounting holes

Two M3 screw holes for easy panel integration.

Key Applications

Automotive systems, Robotics, Home automation, IoT devices

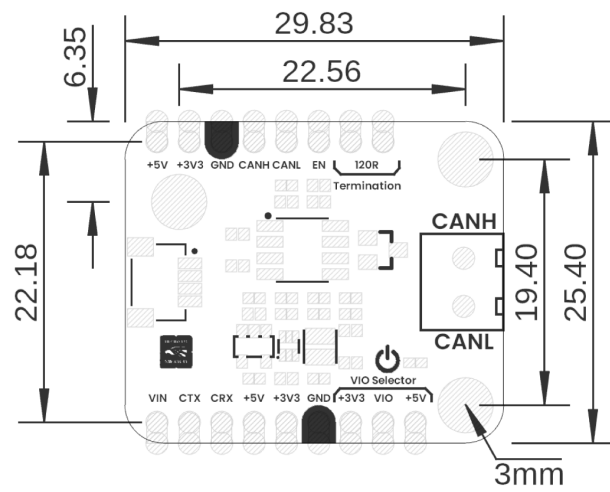
ADDITIONAL TECHNICAL INFORMATION

OVERVIEW

FEATURE / COMPONENT	DETAILS
Communication	High-speed CAN interface for reliable data transfer.
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IC1 – TCAN1051HVD	Automotive-grade CAN transceiver IC.
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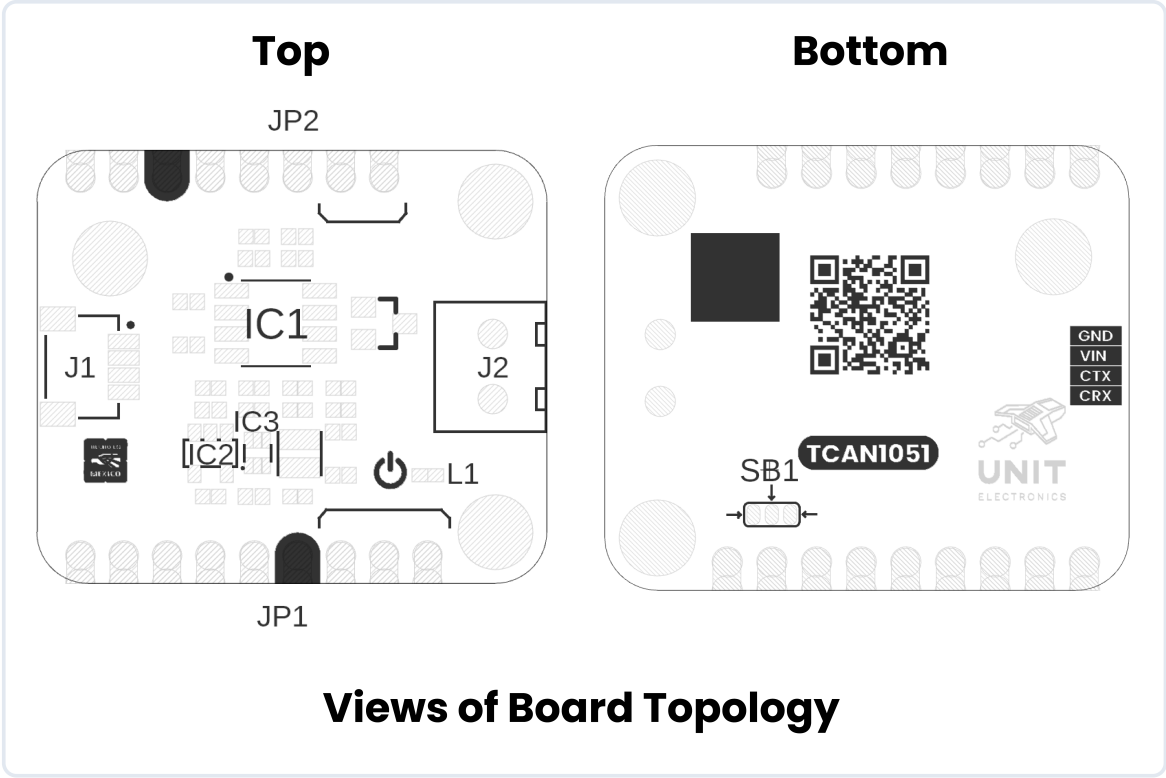
HARDWARE DOCUMENTATION

MECHANICAL DIMENSIONS

**Board Dimensions in Millimeters**

Physical dimensions and mounting specifications (measurements in millimeters)

SYSTEM TOPOLOGY



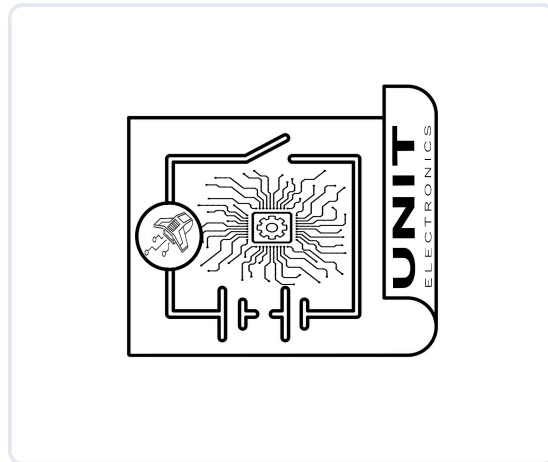
Connection topology and system integration diagram

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SB1	Solder Bridge for VIO

CIRCUIT SCHEMATIC



Complete circuit schematic showing all component connections

[View Complete Schematic PDF](#)

PIN DESCRIPTION

Detailed pin assignment and electrical specifications

SIGNAL DESCRIPTION

FUNCTION		NOTES
Power Supply	3.3V or 5V	
Ground	Common ground for all components	

PIN CONFIGURATION LAYOUT

Physical connector layout and pin positioning



Pin Configuration Layout

Complete pin configuration diagram showing all connectors, pin assignments, and electrical connections for proper integration

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Professional Technical Datasheet

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